Monielle Alencar

Sewall Conservation Ecology Lab, Temple University/Ambler Field Station
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Education

Temple University, Department of Biology - Doctor of Philosophy in *Biology* **2026 – Present**

Federal University of Maranhao, Department of Biology - Master of Science in *Biodiversity and Conservation* **2014 – 2026**

Federal University of Maranhao, Department of Oceanography and Limnology - Bachelor of Science in *Oceanography* **2009 - 2013**

Research Experience

Temple University, Philadelphia, Pennsylvania **2026**
Graduate Researcher, Sewall Conservation Ecology Lab
- Developing dissertation research examining how functional traits and pos-disturbances management strategies shape vegetation dynamics and recover across ecosystems.

Federal University of Maranhao, Brazil **2026**
Research Associate, LEB-Ufma
- Co-mentor for undergraduate students, support is provided in the culmination of their thesis.
- Responsibilities encompass the composition of grant proposals and the active pursuit of sponsorships for research projects and scientific gatherings.

Amares Institute, Maranhao, Brazil **2018 – 2021**
Researcher Associate
- Conducted research and conservation efforts focused on coastal and marine ecosystems, with a specific focus on the rehabilitation, reintroduction, and protection of marine wildlife.
- Engaged in education initiatives aimed at fostering sustainable marine resources utilization within coastal communities, contributing to the advancement of educational policies in this regard.

Federal University of Maranhao, Brazil, Department of Biology **2014 – 2016**
Graduate Research Assistant, Hood Lab
- Analyzed how surrounding ecosystems and soil nutrients influence coastal vegetation, providing insight applicable to biodiversity management in urban-influenced landscapes.

Federal University of Maranhao, Brazil, Department of Biology **2014**
Community Outreach Project, LEB-Ufma
- I collaborated in the design and implementation of a sensory garden during Brazil's National Science and Technology Week, guiding visitors through multisensory activities with plants. I promoted awareness of plant diversity, conservation, and the role of academic research in botany.

Federal University of Maranhao, Brazil, Department of Oceanography and Limnology **2013 – 2014**
Undergraduate Sponsored student, Urbano's Lab

- Described and catalogued plant species of interest and use by the riverside communities of the region and characterized by literature and fieldwork, the main species in their morphological, physiological, and environmental aspects as well as its potential for reforestation projects in human-modified areas.

Federal University of Maranhao, Brazil, Department of Oceanography and Limnology **2012 – 2013**
Undergraduate Sponsored student, Urbano's Lab

- Conducted floristic and phytosociological sampling through observation of the areas suffered some damage such burning, brushed and deforestation, and determined if that affected colonization patterns.

Federal University of Maranhao, Brazil, Department of Oceanography and Limnology **2011 – 2013**
Undergraduate Sponsored student, Urbano's Lab

- Linked soil chemistry and vegetation parameters to environmental gradients to analyzing environmental changes in urban ecosystems.

Federal University of Maranhao, Brazil, Department of Oceanography and Limnology **2011 – 2012**
Undergraduate Sponsored student, Urbano's Lab

- Monitored plant species conservation status on Baixada Maranhense to determine nursery conditions in seedlings productions stage of specimens.

Teaching Experience

Graduate Diploma Coordinator - Unyleya Educational, Brazil. Distance Learning **2020 – Present**

- Coordination of two academic courses in high education (Management and Conservation of Wild and Exotic Fauna, and Management and Control of Bio-Invasive Species), including planning, content creation, quality assurance and team mentorship to ensure positive learning outcomes.

Course Instructor - Unyleya Educational, Brazil. Distance Learning **2020 – Present**

- Ecology of Coastal and Marine Environments – 4 credits
- Prevention, Early Detection, and Rapid Response for Invasive Alien Species in Protected Areas – 4 credits
- Ecology of Invasive Species – 4 credits
- Main Vector and Strategies for Monitoring and Prevention of Invasions – 3 credits
- Commercial Breeding of Wildlife – 4 credits
- Ecology of Wildlife – 4 credits

Online Tutor Fellow Education - Brazilian Federal Agency for Support and Evaluation of Graduate (CAPS), Brazil, Distance Learning **2022**

- As an online tutor in the CAPS C10 course, I guided and supported 30 students in their learning journey.
- Responsibilities included facilitating discussions, providing individualized support, assessing assignments, and conducting Q&A sessions. This experience enhanced my online communication skills, adaptability, and ability to effectively support a diverse group of students.

Federal University of Maranhao, Brazil – Systematic of Phanerogams **2015**

- Leads via distance learning students residing in rural areas in the Northeast of Brazil. Conducted monthly in person sessions that facilitated the construction of knowledge.

Federal University of Maranhao, Brazil – Anatomy and Morphology of Vascular Plants **2014**

- Supported undergraduate learning in taxonomy and organizing lab sessions, grade test.

Publications

Publication names: Machado, M. A. (previously published as); Alencar, M. (current name).

Papers

M.A, Machado, E.B, Almeida Jr, 2019. Spatial structure, diversity, and edaphic factors of an area of Amazonian coast vegetation in Brazil. Journal of the Torrey Botanical Society, 146, 58–68.

B.E.F. Correia, E.B, Almeida Jr**, M.A, Machado**, 2020. Lista floristica e formas de vida da vegetacao de uma restinga em Alcantara, litoral ocidental do Maranhao, Nordeste do Brasil*. Revista Brasileira de Geografia Física (in Portuguese)*

E.B, Almeida Jr**, M.A, Machado**, 2017. Checklist of the restingas of Maranhao State, Northeast, Brazil. Indian Journal of Applied Research, 7, 603–612

E.B, Almeida Jr**, M.A, Machado**, 2016. Floristic of vegetation with marine influence in the south coast of Alagoas, Northeast, Brazil. Indian Journal of Applied Research, 9 (5)

Book Chapters

Almeida JR, E. B., Amorim, I. F. F., **Machado. M, A.**, Correia, B. E. F. Lacerda, D. M. A., Santos-Filho, F. S. A economia do babaçu: das quebradeiras de coco às aplicações biotecnológicas. In: Maciel, A. P. (Org.). Biocombustíveis de babaçu: ensaio técnico sobre oportunidades de produção de biocombustíveis a partir do coco de babaçu. 1ed.São Luis, 2016, v., p. 59-81. (in Portuguese)

Pinheiro, C. U.B; **Machado. M, A**. Identificacao e Caracterizacao de Grupos Ecofisiologicos de Vegetais de Ambientes Ribeirinhos da Baixada Maranhense para Uso em Modelos de Recuperacao e Conservacao de Matas Ciliares na Regiao. In: Claudio Urbano B. Pinheiro. (Org.). Matas Ciliares: Recupercao e Conservacao em Áreas Úmidas do Maranhao. 1ed. Sao Luis: Aquarela, 2013, v. 1, p. 12-250. (in Portuguese)

Fellowship

Simons Graduate Fellowship in Ecology and Evolution – Simons Foundation **2026 -2029**
Competitive, fully funded fellowship supporting three years of doctoral research in the United States beginning Fall 2026. This award recognizes outstanding early-career researchers with exceptional potential to advance fundamental understanding in ecology and evolution, providing full tuition, fees, and stipend support for innovative PhD research.

World' Citizen Program – Maranhao Foundation of Support to Research **2017**
Exchange program focused on academic training in specimen collection at New York Botanical Garden.

Plus-Qualification Program – Maranhao Foundation of Support to Research **2013**
Fully funded fellowship at Rio de Janeiro Herbarium, specializing in training for ethnobotanical survey techniques. Focused on studying species of socio-economic interest in the coastal zone of Brazil, this experience has enriched my expertise in connecting biodiversity with community-based applications.

Work Experience

Oceanographer – CEPEMAR Environmental Solutions **2019**

- Participated in marine mammal monitoring for an environmental study supporting the expansion of port areas in Sao Luis, Maranhao, Brazil, as part of the Environmental Licensing process.
- Gained experience in applied environmental assessments for coastal urban development projects.

Wildlife and Environmental Supervisor – Secretary of State and Natural Resources, Brazil **2017 - 2019**

- Proficient in process analysis, ensuring compliance with wildlife project licensing procedures. I also provide expert technical advice on wildlife-related activities and environmental impact assessment.
- Coordinated and managed the Interinstitutional Working Group for controlling harmful exotic fauna in urban areas and contributed to the development of state wildlife protection regulations Decree No. 34.324 of July 12, 2018.
- Co-led the development of the Management Plan for Rangedor State Park, balancing conservation goals with urban pressures.
- Focal point for Maranhão in the GEF Pró-Espécies Project ("All Against Extinction"), I participated in national discussions in Brasília (Brazil's capital), conferences, and training programs, while compiling biodiversity data on the state's fauna and flora and identifying priority areas for conservation.